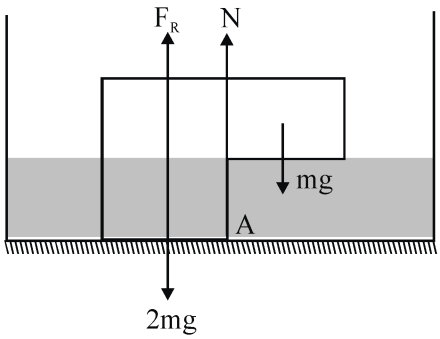
SOLUTION
SECTION-I

Single Correct Answer Type **9 Q. [3 M (–1)]**

1. Ans. (A)

Sol.



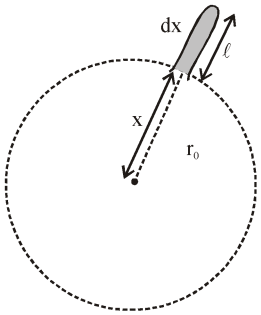
FBD when the material just topples,

Balancing torque, about A : $\left(-F_B+2mg\right)\frac{a}{2}=mg.\frac{a}{2}\Rightarrow F_B=mg\Rightarrow \rho_S=\rho_W$

If $\rho_S<\rho_W$, substance will topple clockwise.
If $\rho_S>\rho_W$, substance will not topple & normal will shift to balance the torque by F_B

2. Ans. (A)

Sol. For the section of length dx,
$dT = \rho A\, dx\, \omega^2 x$



$$\therefore \text{ T at centre } = \int_{r_0}^{\left(\frac{\ell}{2}+r_0\right)} \rho A\omega^2 x\,dx = \frac{1}{2}\rho A\omega^2\left[\left(\frac{\ell}{2}+r_0\right)^2-r_0^2\right]$$

$$= \frac{1}{2}\rho A\omega^2\left[\frac{\ell^2}{4}+r_0^2+\ell r_0-r_0^2\right] = \frac{1}{2}\rho A\omega^2\left[\frac{\ell^2+4\ell r_0}{4}\right] = \frac{1}{8}\rho A\omega^2\ell\left[\ell+4r_0\right]$$

$$\therefore \text{ Pressure } = \frac{T}{A} = \frac{1}{8}\rho\omega^2\ell\left[\ell+4r_0\right]$$

3. Ans. (A)

Sol. (i) $V_{ice} > V_{water}$ so upper layer falls

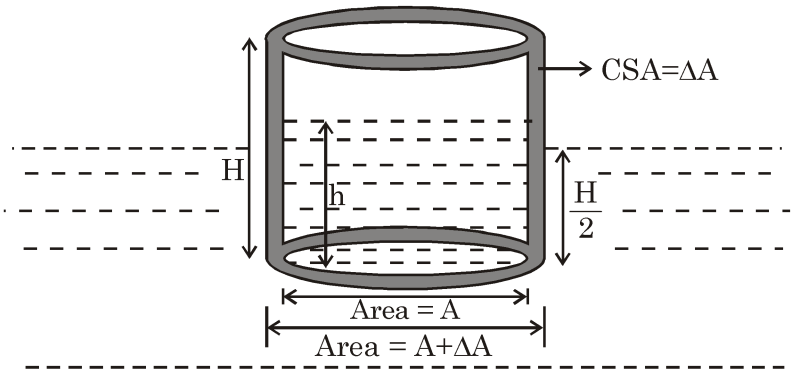
(ii) Displaced volume of water $= \left[M_{ice}-\frac{F_B}{g}\right]\frac{1}{\rho_{water}}$

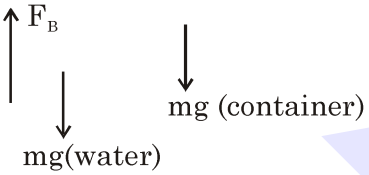
ice melts then B decreases then level of water rises.

For More Material Join: @JEEAdvanced_2024

4. Ans. (A)

Sol.



For equilibrium $\Rightarrow$ FBD $\Rightarrow$ 

$$F_B = mg_{(Water)} + mg_{(container)}$$

$$\rho_W (A + \Delta A)\frac{H}{2}g = \rho_W hAg + \rho_C \rho_W H\Delta Ag$$

$$hA = \left(\frac{AH}{2}\right) + \left(\frac{\Delta AH}{2}\right) - (\rho_C \Delta AH)$$

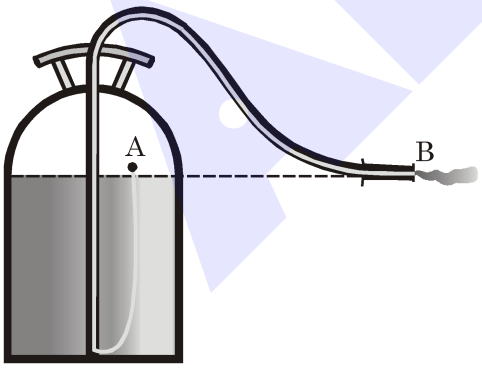
$$h = \frac{H}{2} + \frac{H\Delta A}{A}(0.5 - \rho_C)$$

This shows that $h > \frac{H}{2}$ if $\rho_C < 0.5$

So Ans. (A)

5. Ans. (B)

Sol.



Writing Bernoulli's equation along flow line at point A & B

$$P_{tank} + P_0 + 0 + \frac{1}{2}\rho V_A^2 = P_0 + 0 + \frac{1}{2}\rho V_B^2$$

$$\Rightarrow V_B = \sqrt{\frac{2P_{tank}}{\rho}}$$

[$V_A << V_B$ as area of nozzle is very less as compared to area of tank]

For More Material Join: @JEEAdvanced_2024

6. Ans. (C)

Sol. As the height decreases, the rate of flow with which the water is coming out decreases.

7. Ans. (C)

Sol. Pressure at point inside tap is less than P_0

8. Ans. (B)

Sol. $F = \frac{\Delta p}{\Delta t} = \frac{m(v_1 + v_2)}{\Delta t} = V\rho(v_1 + v_2)$ where $\frac{m}{\Delta t} = V$ = volume/second.

9. Ans. (A)

Sol. Let y be the height of liquid at some instant.

Then $-\frac{dy}{dt}$ = constant (given)

From equation of continuity,

$(\pi x^2)\left(-\frac{dy}{dt}\right) = a\sqrt{2gy}$ (a = area of hole)

Here, $\pi\left(\frac{-dy}{dt}\right)$, a and g are constants

Hence, squaring the equation, we get $y = kx^4$.

Multiple Correct Answer Type 4 Q. [4 M (–1)]

10. Ans. (B, C)

Sol. $F_H = (\rho gR)\pi R^2 = \rho g\pi R^3 = \frac{3Mg}{2}$

$Fv = (M + m)g$

$F_{net} = \sqrt{[(M+m)g]^2 + \left(\frac{3Mg}{2}\right)^2}$

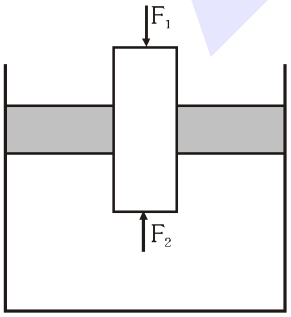
$F_{net} = Mg\sqrt{\frac{9}{4}\left(1 + \frac{m}{M}\right)^2}$

11. Ans. (B,C,D)

Sol. $P_{new} = \rho(g + \frac{g}{2})\,h$ & $B_{new} = P_{new} \times V$

12. Ans. (A,D)

Sol.



$F_2 - F_1$ = upthrust

$\therefore F_2 >$ upthrust

For More Material Join: @JEEAdvanced_2024

13. Ans. (A,C)

Sol. $\rho V_A g = (m_{ice} + m_A) g$ and $\rho V_B g = (m_{ice} + m_B) g \Rightarrow \frac{V_A}{V_B} = \frac{m_{ice} + m_A}{m_{ice} + m_B} = \frac{0.9 \times 980 + 4.9 \times 20}{0.9 \times 980 + 1.9 \times 20} = \frac{49}{46}$

Also A will start sinking first.

Linked Comprehension Type (1 Para × 2Q, 1 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

14. Ans. (D)

Sol. Force exerted by liquid = (mass of liquid displaced) × $\sqrt{g^2 + (\omega^2 r)^2}$

$= \rho_w v \times \sqrt{g^2 + (\omega^2 r)^2}$

15. Ans. (B)

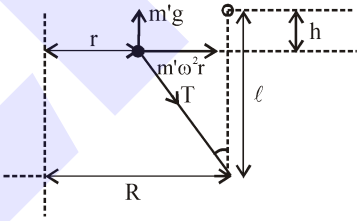
Sol. $R - r = \ell \sin\theta$ ..... (i)

$\ell - h = \ell \cos\theta$ ..... (ii)

$m' = (\rho_w - \rho_{body})v$ & $m'g = T\cos\theta$ ..... (iii)

$m'\omega^2 r = T \sin\theta$ ..... (iv)

from (i), (ii), (iii) & (iv)



$$\frac{\omega^2 r}{g} = \tan\theta = \frac{R - r}{\ell - h}$$

16. Ans. (C)

Sol. Net force exerted by liquid on styrofoam is buoyant force = $\rho_W Vg = \rho_W(A)\,Lg = \rho_w g \frac{\ell^2 L}{2}$

17. Ans. (D)

Sol. Average pressure on slant surface $P_{avg} = \frac{(P_0 + \rho_w g h_0) + \left(P_0 + \rho_w g h_0 - \frac{l}{\sqrt{2}}\right)}{2} = \left(P_0 + \rho_w g h_0 - \frac{l}{2\sqrt{2}}\right)$

Force on any one of the slant face = $\left[P_0 + \rho_w g(h_0 - \frac{\ell}{2\sqrt{2}})\right] L\ell$

18. Ans. (B)

Sol. Balancing force in vertical

$(M_{styrofoam} + m)g + F_{surface\ Tension} = F_{bouyancy}$

$$\rho_f \left(\frac{1}{2}\ell^2 L\right) g + mg + T\left(2L\sin 45° + \frac{2\ell}{\sqrt{2}}\right) = \rho_W \frac{1}{2}\left(\frac{\ell}{\sqrt{2}} + \sqrt{2}\ell\right)\left(\frac{\ell}{\sqrt{2}} - \frac{\ell}{2\sqrt{2}}\right) Lg$$

$$mg = \frac{-\rho_f g L \ell^2}{2} + \frac{3}{8}\rho_w g L \ell^2 - \sqrt{2}T[L + \ell]$$

For More Material Join: @JEEAdvanced_2024

SECTION-II

Numerical Answer Type Question **6 Q. [3(0)]**
(upto second decimal place)

19. Ans. 0.1

Sol. $\rho_1 g h_1 = \rho_2 g h_2$

$$\frac{h_2}{h_1} = \frac{\rho_1}{\rho_2} = \frac{\rho_1}{\rho_1(1-\gamma\Delta T)}$$

$Y = 10^{-5}/°C$, $\Delta T = 100°C$, $h_1 = 300$ cm

$h_2 - h_1 = 0.1$ cm ans.

20. Ans. 4.5

Sol. $P_\ell V_\ell = P_f V_f$

$$P_0(V) = P_f\left(\frac{2V}{3}\right)$$

$$P_f = \frac{3}{2}P_0 = P_0 + \rho_w g h$$

$$\frac{P_0}{2} = \rho_w g h$$

$$\frac{\rho_w g \times 9}{2} = \rho_w g h$$

[h = 4.5 m]

21. Ans. 100

Sol. $T_{max} = 7 \times 10^6 \times 10^{-6} N = 7N$

$\therefore F_{B_{min}} = (15-7) = 8N$

$\therefore$ Water level its be lowered = 2 cm

$\therefore$ Area occupied by water level at the top = 100 cm^2

$\therefore$ $2t = 100 \times 2 \Rightarrow t = 100$ sec

22. Ans. 110

Sol. $I\alpha = (V\rho g)\frac{\ell}{4}$

where $V = \pi r^2 \ell$

so $\alpha = \frac{\pi r^2 \rho g \ell^2}{4I} = 35\pi = 110$ rad/sec^2

23. Ans. 2500

Sol. $A_1V_1 = A_2V_2$

$10 \times 1 = 5\ V_2$

$V_2 = 2$ m/s

$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2$$

$$4000 + \frac{1}{2} \times 10^3 (1)^2 = P_2 + \frac{1}{2} \times 1000 (2)^2$$

$P_2 = 2500$

For More Material Join: @JEEAdvanced_2024

24. Ans. 40

Sol. $mg = \rho_w \frac{v}{3} g = \rho_A \frac{v}{2} g$ ...(i)

$$\rho_{mixture} = \frac{\rho_w v_0 + \rho_A v_0}{2v_0} = \frac{\rho_w + \rho_A}{2}$$

$$mg = \left[\frac{\rho_w + \rho_A}{2}\right] \times \left[\frac{v}{n}\right] g \text{ ...(ii)}$$

From equation (i) & (ii) $\rho_w \frac{v}{3} g = \frac{vg}{2n}\left[\rho_w + \frac{2\rho_w}{3}\right]; \frac{1}{3} = \frac{1}{2n}\left[\frac{5}{3}\right] \Rightarrow n = \frac{5}{2}$

SECTION-III

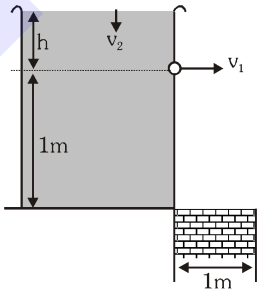
Numerical Grid Type (Ranging from 0 to 9) **3 Q. [4 M (0)]**

25. Ans. 5

Sol For minimum velocity at orifice

$$\Rightarrow (v_{min}) \times \sqrt{\frac{2h}{g}} = 1 \Rightarrow v_{min} = \sqrt{5} \text{ m/s}$$

This will give us minimum height for this velocity



$$\Rightarrow \sqrt{2gh_{min}} = \sqrt{5} \Rightarrow h_{min} = \frac{1}{4} m = 0.25m$$

$v_1 = \sqrt{2gh}$ & $v_2 = -dh/dt$

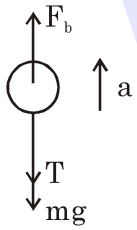
By equation of continuity $av_1 = Av_2$

$$\Rightarrow a\sqrt{2gh} = A\left(-\frac{dh}{dt}\right) \Rightarrow \int_{0.81}^{0.25} \frac{dh}{\sqrt{h}} = \int_0^t \frac{\left(a\sqrt{2g}\right)}{A} dt$$

$\Rightarrow t = 125 \text{ sec} = (5)^3 = (\alpha)^3 \Rightarrow \alpha = 5$

26. Ans. 3

Sol.



$F_b - T - mg = ma$

$F_b - mg - ma = T$

$4 \times 10^{-3} \times 10^3 \times (10 + 2) - 2 \times 10 - 2 \times 2 = T$

$T = 24$

$$V = \frac{m}{\rho} = \frac{2}{0.5 \times 10^3}$$

$V = 4 \times 10^{-3}$

27. Ans. 2

Sol. Equation of continuity $\Rightarrow v_A + v_B = v_C + v_D$

$\Rightarrow v_D = 2m/s$

For More Material Join: @JEEAdvanced_2024

SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

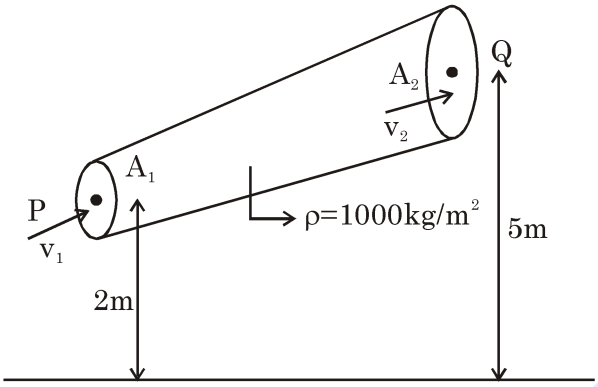
28. Ans. (A) PQ (B) PQ (C) PRST (D) PRST

Sol. If $dgh_1 > P \Rightarrow$ height of liquid will not cross h_1 $dgh_1 < P \Rightarrow$ height of liquid will cross h_1

Subjective Type **2 Q. [4 M (0)]**

29. Ans. + 29625 J/m^3, – 30000 J/m^3

Sol.



$A_1 = 4 \times 10^{-3}$ m^2

$A_2 = 8 \times 10^{-3}$ m^2

$v_1 = 1$ m/sec

$\rho_W = 10^3$ kg/m^3

Continuity equation at P & Q

$A_1v_1 = A_2v_2$

$(4 \times 10^{-3})\,(1) = (8 \times 10^{-3})\,(v_2)$

$$v_2 = \frac{1}{2}\,m/sec$$

Work done per unit volume by pressure = ΔP = pressure difference

Applying Bernoulli's theorem between P & Q

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

$$P_1 - P_2 = \rho g\left(h_2 - h_1\right) + \frac{\rho}{2}\left(v_2^2 - v_1^2\right) = 10^4\left(5-2\right) + 500\left(\frac{1}{4} - 1\right)$$

$$= 30000 - \frac{1500}{4} = 29625\ J/m^3$$

Work done per unit volume by gravity force = $\dfrac{mgh}{v} = (\rho gh)$

So, work done/volume = $\rho g\,(h_1 - h_2)$

$= 1000 \times 10\,[2 - 5]$

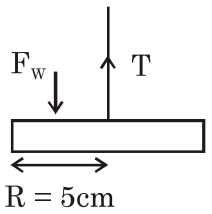
$= -30000$ J/m^3

For More Material Join: @JEEAdvanced_2024

30. **Ans.** $h_1 = \frac{2(3+\pi)}{15\pi} = 0.26$; $h_2 = \frac{3+\pi}{10\pi} = 0.195$

Sol. Plug opens when level of water reaches 'h'.

- FBD of plug



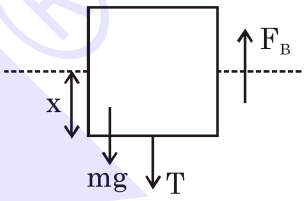
$F_W = T$

$P.\pi R^2 = T$

$$T = \rho_W ghA = 1000 \times 10 \times h \times \pi\left(\frac{5}{100}\right)^2$$

$T = 25\ \pi h$

- FBD of float



$F_B = mg + T$

$\rho_W gxA = 3 \times 10 + T$

$$10000x\pi \times \left(\frac{10}{100}\right)^2 = 30 + 25\ \pi h$$

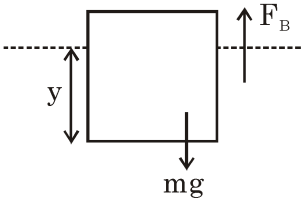
$100\ \pi x = 30 + 25\ \pi h$

$100\pi\ (h - 0.1) = 30 + 25\ \pi h$

$75\ \pi h = 30 + 10\ \pi$

$$h = \frac{(3+\pi)10}{75\pi} = \left(\frac{2(3+\pi)}{15\pi}\right) m$$

- To again close the plug



$mg = F_B$ (for the float) $\rightarrow$ [When water has risen to height y for the float]

$F_B = mg$

$\rho_W gyA = 3g$

$$1000 \times 10 \times y \times \pi \times \left(\frac{10}{100}\right)^2 = 3g$$

$100\pi y = 3g$

$$y = \frac{30}{100\pi} = \frac{3}{10\pi} m$$

$$\text{So } h = 0.1\ m + \frac{3}{10\pi} m$$

$$= \left(\frac{1}{10} + \frac{3}{10\pi}\right) m$$

$$h = \left(\frac{\pi+3}{10\pi}\right) m$$

For More Material Join: @JEEAdvanced_2024